WEARABLE GLOVE– SMART COMMUNICATION DEVICE

BECE304L-
Analog Communication system
Course based Project

Bachelor of Technology in Electronics and communication

by

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ABSTRACT:

The **Assistant Glove** is an innovative wearable device designed to facilitate communication through hand gestures. The system utilizes **three flex sensors** to detect specific finger movements, translating them into predefined messages. These messages are then displayed on an **LCD screen** and transmitted to a **smartphone via Bluetooth** (HC-05). The primary objective of this project is to provide an **efficient and wireless** means of communication for individuals with speech or mobility impairments. The integration of **flex sensors, microcontrollers, and Bluetooth technology** ensures real-time processing and transmission of messages, making the glove a practical assistive tool. The project focuses on **low power consumption, ease of use, and portability**, making it a viable solution for hands-free communication in various applications.

MOTIVATION:

Communication is a fundamental aspect of human interaction, yet individuals with **speech or mobility impairment s**often face significant challenges in expressing themselves. Traditional assistive communication devices can be bulky, expensive, or require complex setups, making them inaccessible to many users.

This project aims to **bridge the communication gap** by developing an **affordable, portable, and user-friendly** solution. The **Assistant Glove** leverages **flex sensors and Bluetooth technology** to recognize hand gestures and convert them into meaningful messages displayed on an **LCD screen** and transmitted to a **smartphone**. By eliminating the need for spoken or written communication, this glove empowers individuals to express themselves more effectively, improving their independence and quality of life.

The motivation behind this project stems from the need for an **accessible assistive device** that is **lightweight, real-time, and easy to use**, providing a **seamless communication experience** for individuals with disabilities or those working in environments where verbal communication is limited.

LITERATURE SURVEY:

The development of assistive communication devices has been an active area of research, with various approaches being explored to enhance accessibility and ease of communication. This literature survey highlights existing technologies and research related to gesture-based communication systems, flex sensor-based wearables, and Bluetooth-enabled assistive devices.

1. Gesture-Based Communication Systems

Gesture recognition has been widely studied as an alternative means of communication for individuals with speech disabilities. Research has shown that **sign language gloves** using flex sensors and accelerometers can effectively interpret hand movements into textual or auditory outputs (**Kumar et al., 2020**). Several studies have implemented **machine learning algorithms** to improve gesture classification accuracy (**Singh et al., 2019**). However, these systems often rely on **complex processing units** and **WiFi connectivity**, which may limit portability and increase power consumption.

2. Flex Sensor-Based Wearables

Flex sensors have been extensively used in wearable technology for tracking finger movements. Studies have demonstrated their effectiveness in detecting finger bending angles and converting them into electrical signals (**Patil & Deshpande, 2021**). Previous projects, such as **smart gloves for sign language recognition**, have successfully integrated flex sensors with microcontrollers like **Arduino and ESP32**. However, many implementations focus on **data logging and sign language translation**, rather than real-time LCD display and direct Bluetooth transmission.

3. Bluetooth-Enabled Assistive Devices

Bluetooth-based assistive devices offer a **wireless, low-power solution** for real-time data transmission. Research indicates that **HC-05 and ESP32-Bluetooth modules** provide reliable communication between wearable devices and smartphones (**Sharma et al., 2022**). Compared to WiFi-based systems, Bluetooth consumes **less power** and ensures seamless connectivity even in offline environments. Despite these advantages, prior studies often emphasize **speech synthesis** rather than **text-based display**, which is more versatile in noisy environments or for users with hearing impairments.

4. Existing Gaps and Proposed Solution

While various research efforts have been made in gesture recognition, wearable technology, and Bluetooth-based communication, existing solutions often require **additional processing units, cloud-based services, or complex setups**. This project addresses these limitations by:

- Using **three flex sensors** to simplify gesture recognition while maintaining accuracy.
- Displaying messages on an **LCD screen** for instant visual feedback.
- Sending messages **directly to a smartphone via Bluetooth** without requiring an internet connection.
- Focusing on **real-time, portable, and low-cost** implementation, making the device more practical for daily use.

This literature survey provides a foundation for the design and development of the **Assistant Glove**, ensuring it builds upon previous research while introducing a **simplified yet effective** communication tool.

PROPOSED METHOD:

The **Assistant Glove** is designed to recognise specific finger movements using **three flex sensors**, process the detected gestures using a **microcontroller**, display the corresponding message on an **LCD screen**, and transmit the message to a **smartphone via Bluetooth**. The methodology involves **sensor integration, data processing, message display, and wireless transmission**.

1. System Overview

The proposed system consists of the following main components:

1. **Input Unit:**
    - **Three Flex Sensors**: Placed on the glove to detect finger bending.
    - **Microcontroller (Arduino)**: Processes the sensor values and maps them to predefined messages.
2. **Processing Unit:**
    - The microcontroller reads the **analog values** from the flex sensors.
    - A **threshold-based decision-making algorithm** determines the corresponding gesture.
    - The mapped message is prepared for display and transmission.
3. **Output Unit:**
    - **16x2 LCD Display**: Shows the corresponding message in real-time.
    - **HC-05 Bluetooth Module**: Sends the message to a paired smartphone.

2. Detailed Methodology

Step 1: Sensor Integration and Calibration

- Attach **three flex sensors** to different fingers of the glove.
- Read and calibrate the **sensor resistance changes** when fingers are bent.
- Define **threshold values** for different bending angles to distinguish gestures.

Step 2: Gesture Recognition and Message Mapping

- The **analog values** from the flex sensors are read and processed.

- A **decision-making algorithm** maps each unique sensor pattern to a predefined message.
- Example mapping:
  - **Gesture 1 (Bend Finger 1)** → "WASHROOM"
  - **Gesture 2 (Bend Finger 2)** → “DRINK WATER"
  - **Gesture 3 (Bend Finger 1 & 2)** → "FOOD"

Step 3: LCD Display for Visual Feedback

- The mapped message is sent to a **16x2 LCD display** using the I2C protocol.
- The user gets **instant visual feedback** on the message being sent.

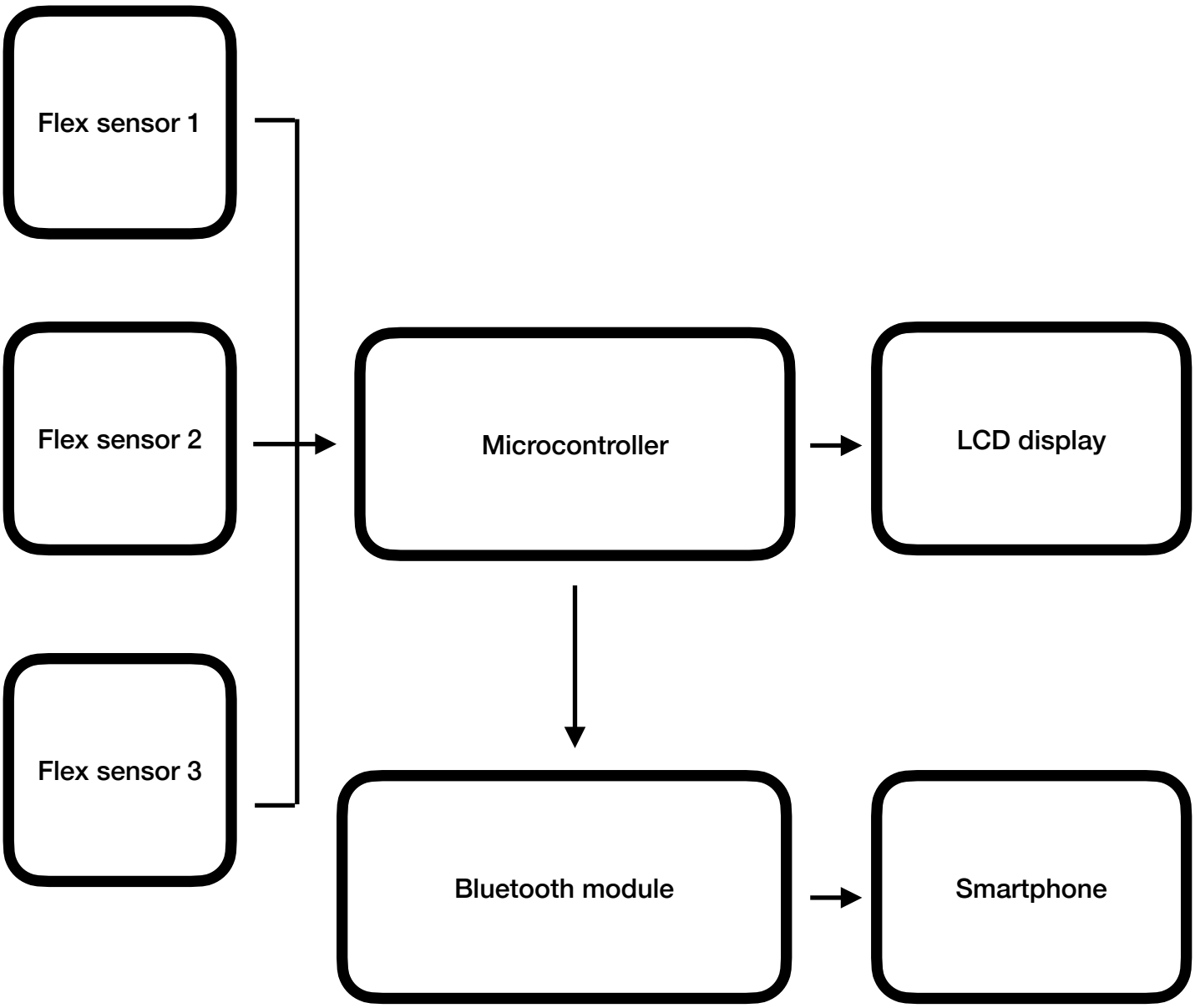
Step 4: Bluetooth Communication with Smartphone

- The **HC-05 (Bluetooth Module) Bluetooth** is initialised.
- The microcontroller transmits the message via **serial communication (UART)**.
- A **paired smartphone** receives and displays the message in a Bluetooth terminal app.

Step 5: Testing and Validation

- Check the accuracy of **gesture recognition** by testing multiple users.
- Verify **LCD display and Bluetooth transmission** under different conditions.
- Ensure **low power consumption** for prolonged usage.

BLOCK DIAGRAM:



CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);  // LCD I2C Address
SoftwareSerial BT(2, 3);  // HC-05 TX → D2, HC-05 RX → D3 (via
voltage divider)

void setup() {
  Serial.begin(9600); // PC Serial Monitor
  BT.begin(9600);     // HC-05 Bluetooth Communication

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("SIGN LANGUAGE");
}

void loop() {
  int s1 = analogRead(A0);
  int s2 = analogRead(A1);
  int s3 = analogRead(A2);

  Serial.print("Flex 1: "); Serial.println(s1);
  Serial.print("Flex 2: "); Serial.println(s2);
  Serial.print("Flex 3: "); Serial.println(s3);
  Serial.println("----------------------");

  lcd.clear();  //  Clears LCD to prevent overlapping text

  String message = "NOTHING";  // Default message

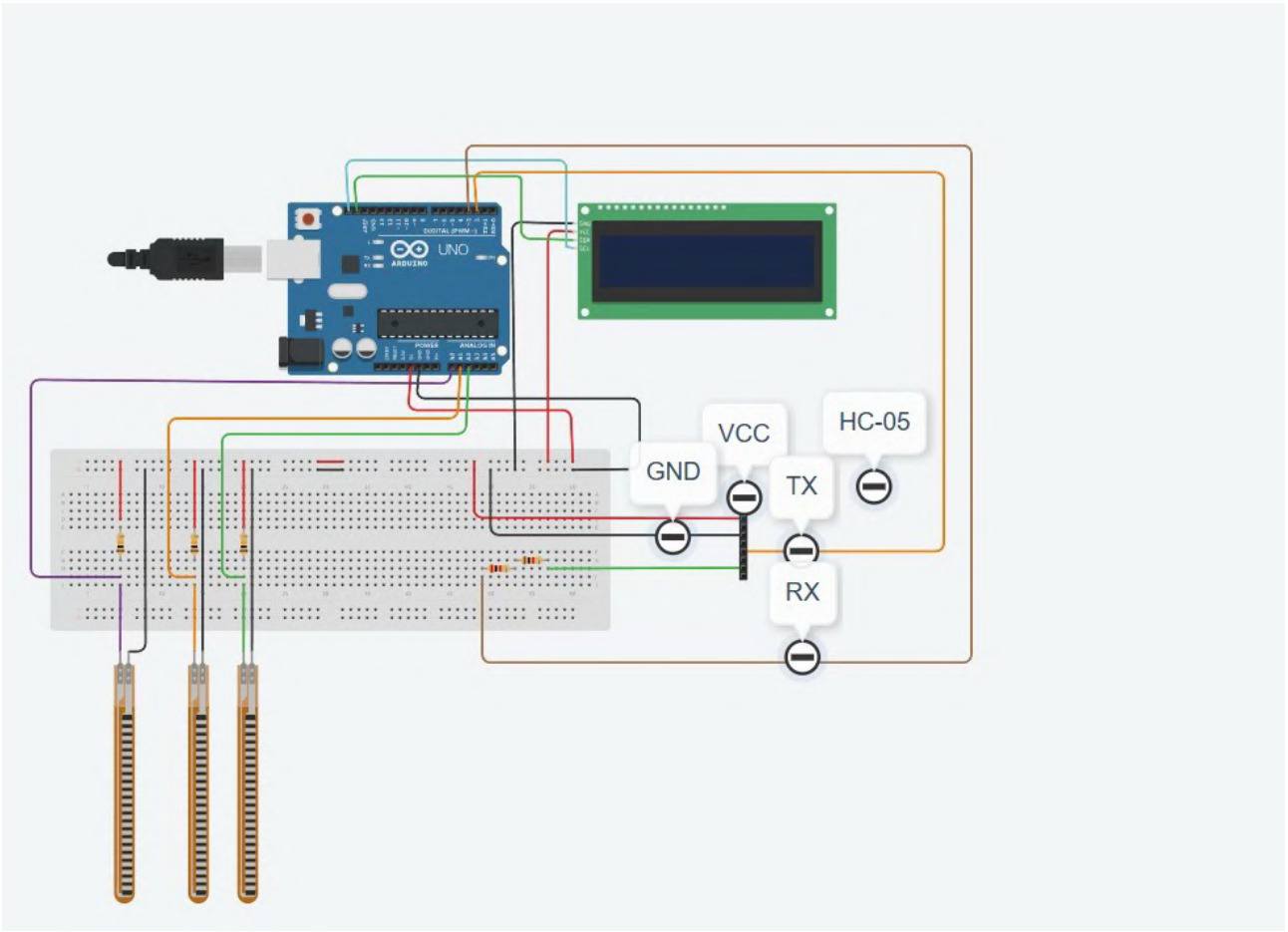
  if (s1 > 200) {
    message = "WASHROOM";
  }
  else if (s2 > 300) {
    message = "DRINK WATER";
  }
  else if (s3 > 700) {
    message = "FOOD";
  }

  lcd.setCursor(0, 1);
  lcd.print(message);  //  Display correct message
  BT.println(message); // Send to Bluetooth
  Serial.println("Sent via Bluetooth: " + message); //

  delay(500);
```

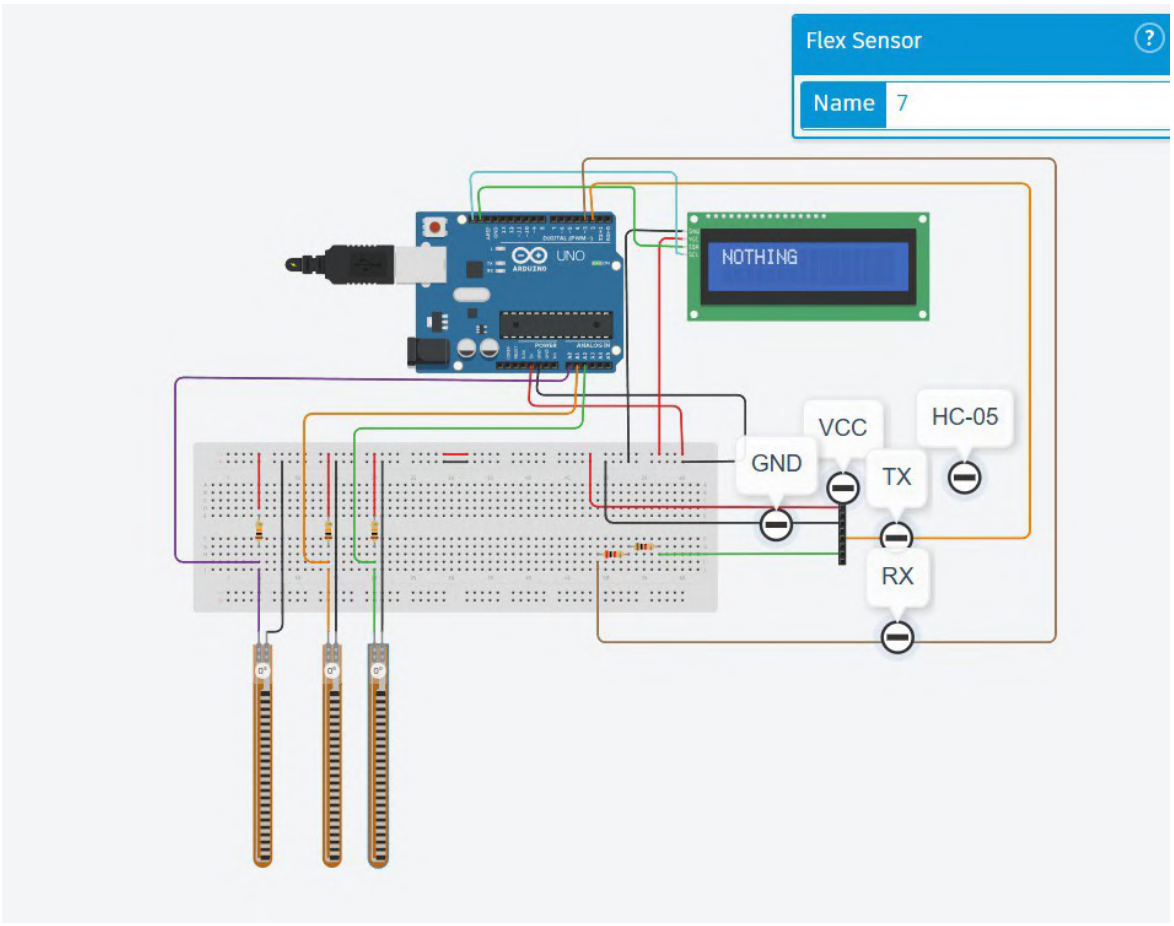
SNAPSHOT OF THE CIRCUIT:

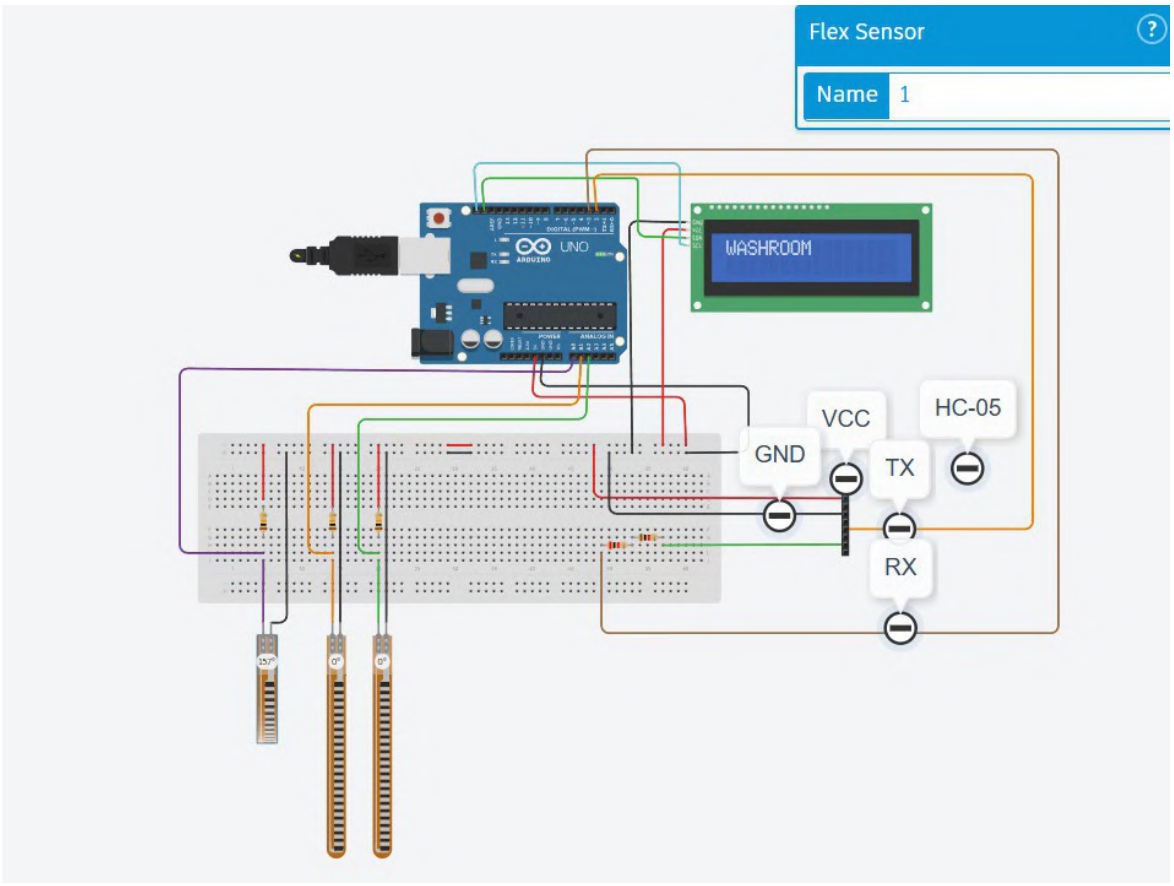


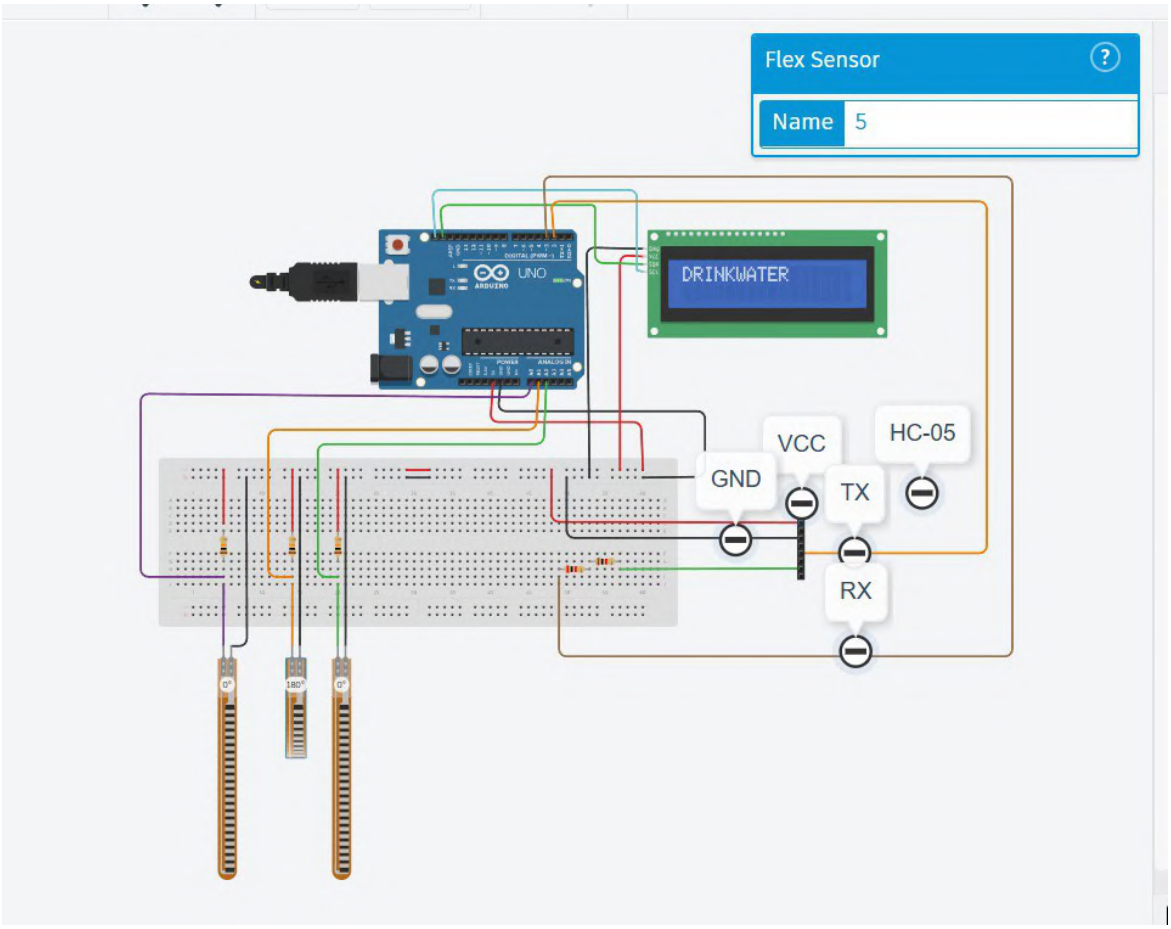


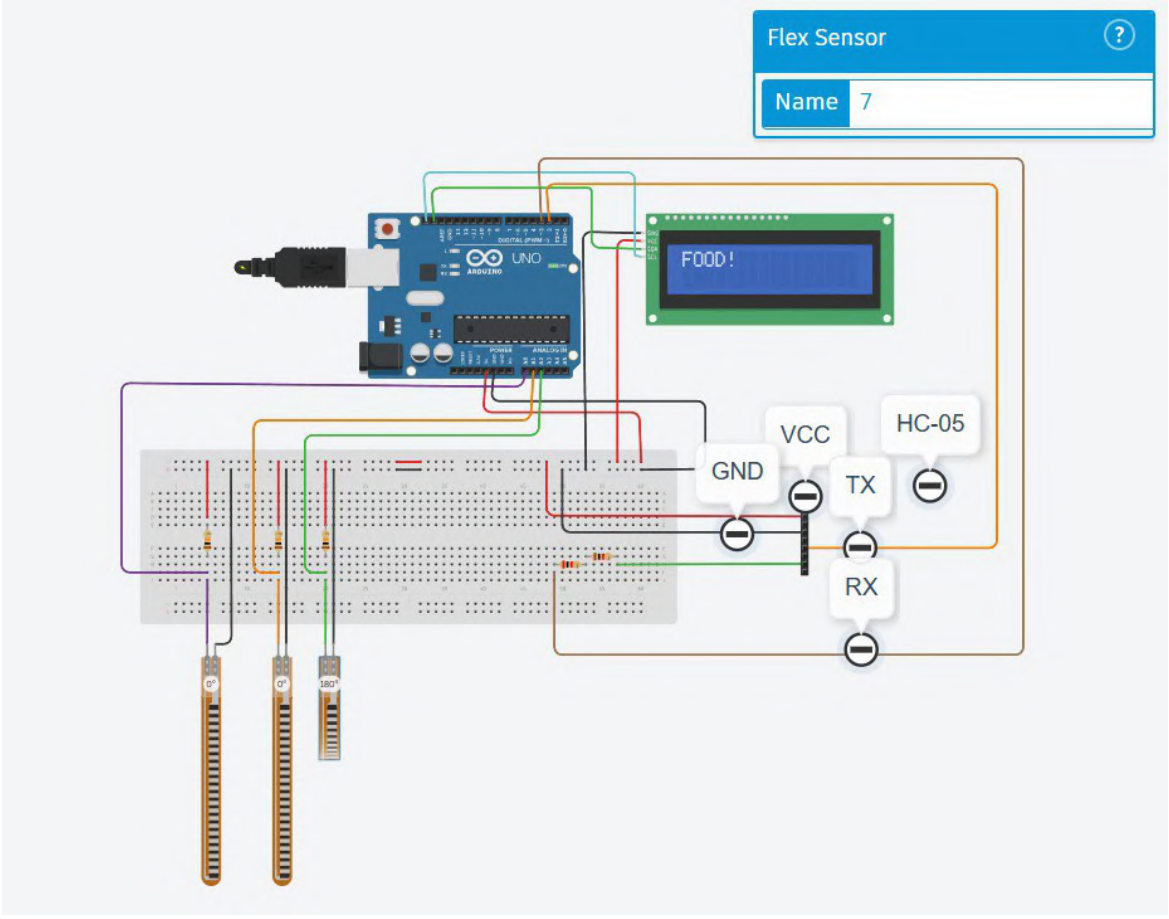
OUTPUT:

SIMULATION OUTPUT:

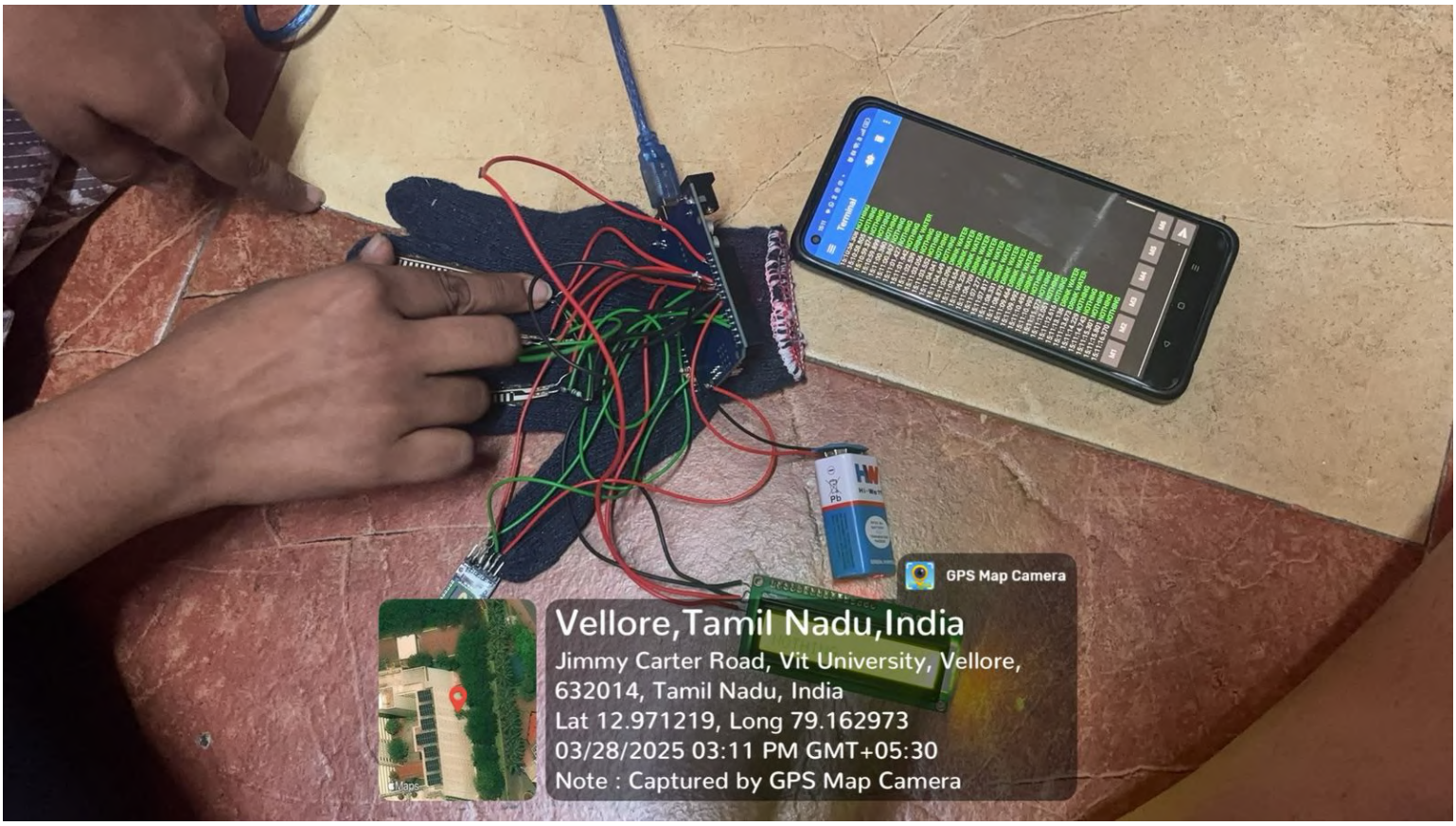


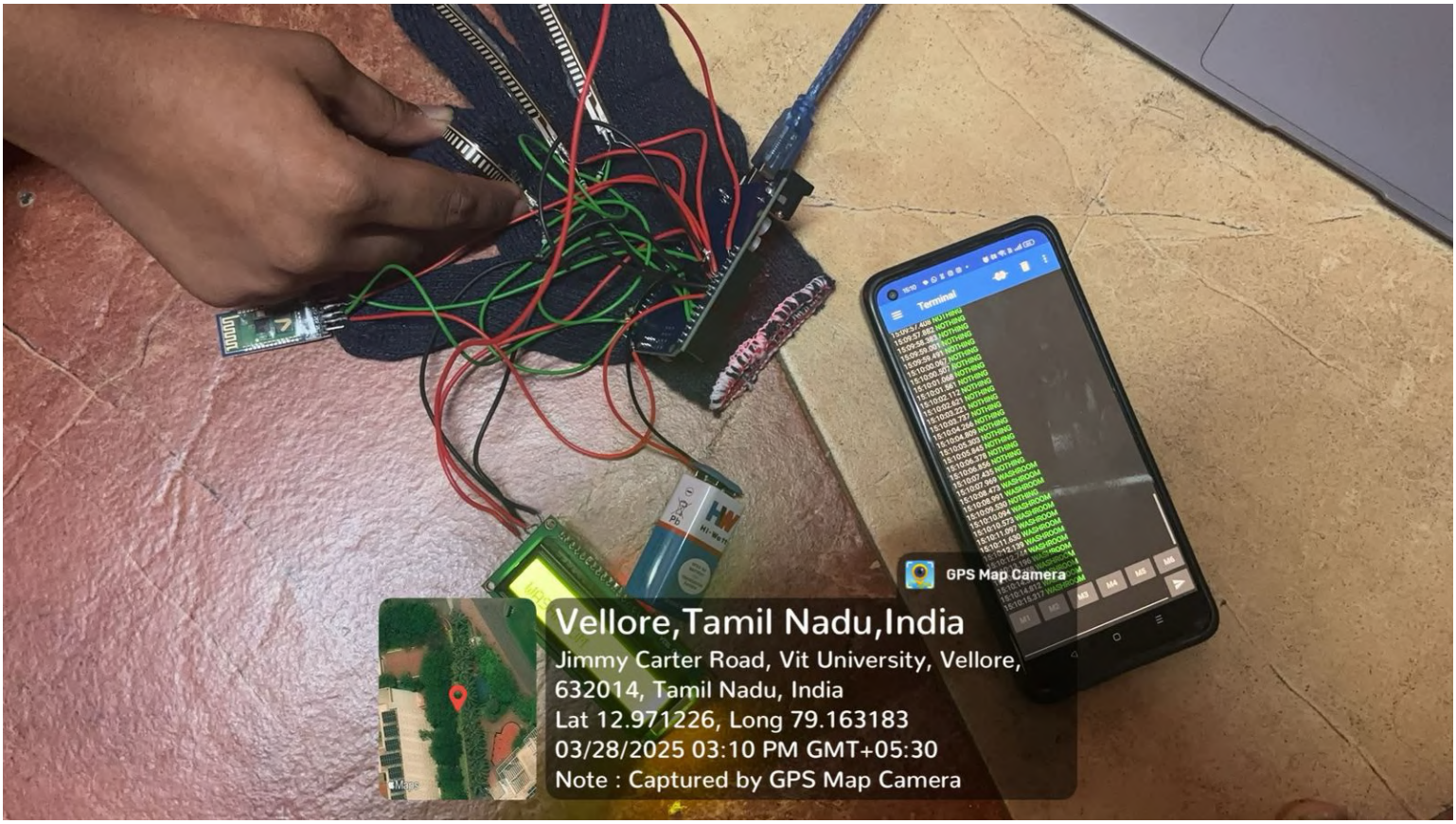



Flex Sensor
Name 5
DRINKWATER
VCC
HC-05
GND
TX
RX

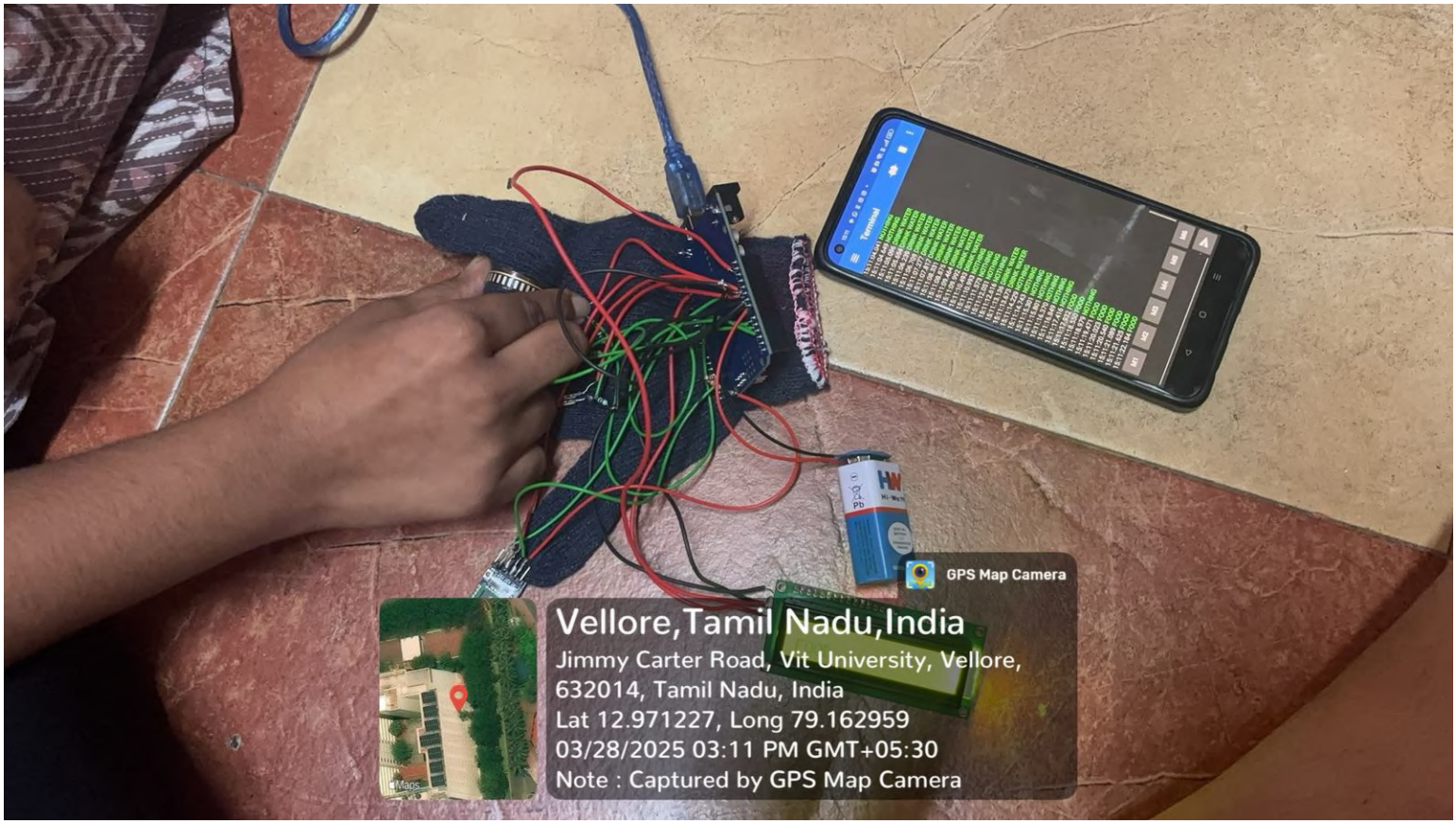

Flex Sensor
Name 7
FOOD!
VCC
HC-05
GND
TX
RX

HARDWARE OUTPUT:






GPS Map Camera
Vellore,Tamil Nadu,India
Jimmy Carter Road, Vit University, Vellore,
632014, Tamil Nadu, India
Lat 12.971265, Long 79.163028
03/28/2025 03:11 PM GMT+05:30
Note : Captured by GPS Map Camera


GPS Map Camera
Vellore,Tamil Nadu,India
Jimmy Carter Road, Vit University, Vellore,
632014, Tamil Nadu, India
Lat 12.971227, Long 79.162959
03/28/2025 03:11 PM GMT+05:30
Note : Captured by GPS Map Camera

EXPLANATION OF THE OUTPUT:

The **Assistant Glove** interprets different finger movements detected by **three flex sensors** and maps them to predefined messages. The recognized messages are displayed on an **LCD screen** and transmitted to a **smartphone via Bluetooth**.

Output Behaviour Based on Flex Sensor Activation:

Flex Sensor Activated	Corresponding Message	Displayed on LCD	Sent to Smartphone
No sensor bent (default)	NOTHING	NOTHING	NOTHING
Flex Sensor 1 bent	WASHROOM	WASHROOM	WASHROOM
Flex Sensor 2 bent	NEED WATER	NEED WATER	NEED WATER
Flex Sensor 3 bent	FOOD	FOOD	FOOD

1. **Default State (No Flex Sensor Bent)**
    - When the hand is at rest, none of the flex sensors are activated.
    - The system detects this as a **default state** and displays **"NOTHING"** on the LCD.
    - The same message is transmitted via Bluetooth.
2. **Flex Sensor 1 Bent → "WASHROOM"**
    - When **Flex Sensor 1** is bent, the system maps it to the message **"WASHROOM"**.
    - The LCD instantly displays **"WASHROOM"** for visual feedback.
    - The message **"WASHROOM"** is sent via Bluetooth to the smartphone.
3. **Flex Sensor 2 Bent → "NEED WATER"**
    - If **Flex Sensor 2** is bent, the system recognizes it as **"NEED WATER"**.
    - The LCD displays **"NEED WATER"** and the same message is sent to the smartphone.
4. **Flex Sensor 3 Bent → "FOOD"**
    - When **Flex Sensor 3** is bent, the message **"FOOD"** is displayed on the LCD.
    - The Bluetooth module transmits **"FOOD"** to the smartphone.

Special Cases & Considerations

- **Multiple Sensors Bent Simultaneously**
  - If two or more flex sensors are bent at the same time, the system can be programmed to prioritise one message or send a **combined message** (e.g., "NEED WATER & FOOD").
- **LCD and Bluetooth Synchronisation**
  - The LCD ensures that the user sees the message before it is transmitted.
  - The Bluetooth transmission allows remote caregivers to receive the request instantly.
- **Practical Use Case**
  - The user can communicate basic needs like requesting **food, water, or access to the washroom** without speaking.
  - This system is **helpful for individuals with speech impairments, elderly users, or those in noisy environments** where verbal communication is difficult.

ENHANCEMENTS THAT CAN BE MADE:

Improved Flex Sensor Accuracy

- Use **capacitive flex sensors** for better sensitivity and durability.
- Implement **sensor calibration** to adapt to different hand sizes and stiffness variations.

Mobile App Integration

- Develop an **Android/iOS app** to display instructions more interactively.
- Add **voice output** to assist visually impaired users.

Power Efficiency Optimisation

- Use **low-power microcontrollers** (e.g., ESP32-Low Power Mode).
- Implement **auto-sleep mode** to extend battery life.

Wireless Connectivity Upgrades

- Replace HC-05 with **BLE (e.g., HM-10 or ESP32 Bluetooth)** for lower power consumption.
- Add **offline storage & sync** for logging gesture history.

Enhanced Wearability & Comfort

- Use **conductive fabric-based flex sensors** for a **softer, more comfortable glove**.
- Integrate a **vibration motor** to provide haptic feedback for confirmation.

AI-Based Gesture Recognition

- Implement **machine learning (ML)** to recognise more complex hand movements.
- Use **IMU sensors (gyroscope + accelerometer)** for **dynamic gestures**.

Adding a heartbeat sensor and temperature sensor to the Assistant Glove can enhance its functionality in several ways:

1. Health Monitoring

- The heartbeat sensor can continuously track the user's heart rate, providing real-time health data, which is especially useful for individuals with medical conditions.

- The temperature sensor can detect abnormal body temperature, which could indicate fever, infection, or other health issues.

2. Emergency Alerts

- If the heartbeat sensor detects an irregular pulse (too high or too low), it can trigger an alert to caregivers or emergency services.

- If the temperature sensor detects an extreme body temperature, it can warn the user or notify a medical professional.

3. Smart Adaptive Control

- The glove could adjust its sensitivity based on physiological data. For example, if the user is fatigued (based on a low heart rate), the glove could increase signal sensitivity to require less effort for interaction.

- Temperature readings can be used to adjust comfort settings, like activating a vibration alert or adjusting material airflow.

4. Enhanced Safety for Patients & Elderly Users

- The glove could help monitor stress levels based on heart rate variability.

- It could be programmed to send alerts to caregivers if a sudden drop in heart rate or a fever is detected.

5. Integration with IoT and Mobile Apps

- The sensors can send real-time data to a smartphone app for tracking and analysis.

- Users or doctors can monitor long-term health trends through cloud-based storage.

APPLICATIONS AND FUTURE SCOPE:

- Healthcare & Hospitals – Helps elderly, disabled, and ICU patients request assistance easily.

- Industrial & Workplace Safety – Workers can signal for help in emergencies without stopping work.

- Restaurants & Hotels – Customers can request service hands-free, improving efficiency.

- Smart Home & Assistive Tech – Elderly users can call for help or integrate with IoT home automation.

- Military & Defence – Soldiers can send silent distress signals in covert missions.

- Public Transport & Travel – Disabled or elderly passengers can request assistance easily.

- Disaster & Rescue Operations – Trapped individuals can send emergency alerts.

- Education & Special Needs – Helps students with disabilities communicate effectively.

Future Scope: IoT & AI-based gesture control, health monitoring, mobile app integration.

CONCLUSIONS:

The Wearable Automatic Assistance Glove is a smart, hands-free communication device that enables users to request help effortlessly. It is highly beneficial in healthcare, industrial safety, hospitality, and emergency response. By integrating sensors, microcontrollers, wireless transmission, and alert systems, it ensures quick and efficient assistance. Future enhancements like IoT, AI-based gesture recognition, and health monitoring can further improve its functionality, making it a valuable tool for patients, workers, and caregivers alike.